STARTING (Continued)

SPECIFICATIONS

STARTER

Starter Motor and Solenoid	
Manufacturer	Mitsubishi
Engine Application	2.4L, 4.0L
Power Rating	2.4L - 1.4 Kilowatt (1.9 Horsepower) 4.0L - 1.4 Kilowatt (1.9 Horsepower)
Voltage	12 Volts
** Number of Permanent Magnets	6
Number of Brushes	4
Drive Type	Planetary Gear Reduction
Free Running Test Voltage	11.2 Volts
Free Running Test Maximum Amperage Draw	90 Amperes
Free Running Test Minimum Speed	2.4L - 24400 rpm 4.0L - 2500 rpm
Solenoid Closing Maximum Voltage Required	7.8 Volts
*Cranking Amperage Draw Test	2.4L - 160 Amperes 4.0L - 160 Amperes
* Test at operating temperature. Cold engine, tight (new) engine, or heavy oil will increase starter amperage draw.	
** The starter is equipped with permanent magnets. Never strike the starter case to attempt to loosen a sticking/ stuck armature as permanent magnets may crack or break.	

TORQUE

DESCRIPTION	N-m	Ft. Lbs.	In. Lbs.
Starter Solenoid B+ Terminal Nut	11.3	-	100
Starter Mounting Bolts (2.4L)	54	40	-
Starter Heat Shield Mounting Bolts	6	-	55
Starter Mounting Bolt (forward facing 4.0L)	41	30	-
Starter Mounting Bolt (rearward facing 4.0L)	48	35	-

STARTER MOTOR

DIAGNOSIS AND TESTING - STARTER MOTOR/SOLENOID

Correct starter motor operation can be confirmed by performing the following free running bench test. This test can only be performed with starter motor removed from vehicle. Refer to starter motor specifications.

CAUTION: Permanent magnets are used in the starter. Permanent magnet starters are highly sensitive to hammering, shocks, external pressure and reverse polarity. This starter motor must never be clamped in a vise by starter field frame. The starter should only be clamped by mounting flange. Do not reverse battery cable connections to starter motor when testing. The permanent magnets may be damaged and starter rendered unserviceable if it is subjected to any of these conditions.